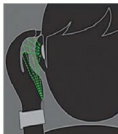


to make the concept possible, the most important being a Bluetooth communication module to provide two-way audio to the phone. A directional speaker is used to fire audio to the user's hand, while an omnidirectional microphone placed beside the speaker reads the user's speech. Noise-cancellation algorithms are included to cancel out the wind and other ambient noises. The watch strap uses a quick-snap magnetic buckle, which provides docking for the Bluetooth audio fob.

Shrinking circuits to fit the mechanical complexity

"The biggest challenge was to shrink the circuits and achieve speaker back volumes in a slim form factor suitable for the wrist," shares Ram Pattikonda, chief technology officer (CTO), iMetrix Technology—the company behind HBand.



The working mechanism of HBand



Ram Pattikonda, the man behind HBand

HBand uses a new strap clasp concept that accommodates the slim fob on the wrist and makes fob-open and pull-out functions easy, without adding to the thickness on the wrist. The electronics and the audio acoustics are miniaturised, while maintaining the style and multi-function capability. The HBand product development took about one-and-a-half years because of the delays in plastic tooling, packaging and integration due to the product's mechanical complexity. The electronics system,

though complex, was designed in less than six months.

Essentials always on your wrist

Almost all smartwatches feature music, call display, weather information and message notification. HBand enhances these features with its audio support and easy-to-use mechanical structure. Its innovative combination of mechanical, electronic and acoustic features takes ordinary smartwatches to another level. The product has already been launched in India and the US.

In fact, there is more to come. The next version of HBand is being designed with focus on better product design and touch-button facilities. The team is also working towards adding features like fitness and sleep tracker, high-resolution OLED display and notification with scrolling text, while extending the battery life to one week. **EFY**

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